A Four-Stage Quantitative Research Pipeline: Signal Discovery, Strategy Construction, Portfolio Construction, and Risk Analysis

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Project Overview

This document summarizes a research pipeline built on daily prices of eight ETFs (SPY, TLT, GLD, USO, UUP, EFA, EEM, FXE) from 2004 to August 2026. It tests whether volatility and market regimes can be modelled, turns those signals and a cross-asset momentum rule into three trading strategies (volatility timing, regime switching and adaptive asset allocation, AAA), combines the strategies into portfolios, and then checks the result with sensitivity analysis, historical stress episodes and a block-bootstrap Monte Carlo. All strategy and portfolio results are walk-forward out of sample (July 2006 to August 2026) and net of transaction costs and slippage. Over this period AAA had a net Sharpe ratio of 0.83 and a maximum drawdown of -17.3%, against 0.65 and -55.2% for SPY buy-and-hold. Adding AAA to a portfolio of the two SPY-based strategies raised the portfolio Sharpe ratio from 0.77 to 0.88 and reduced its maximum drawdown from -23.5% to -16.8% under inverse-volatility weighting, mainly because AAA is much less correlated with the other two strategies. Simple weighting schemes did at least as well as mean–variance optimization. These results come from one historical sample and should be read as research evidence, not as a forecast of future performance.

1 Introduction and Data

The aim of the project is to show a complete, transparent research process rather than to present a production trading system. Each stage asks a separate question: is there a usable signal, does

it survive as a trading rule after costs, does it improve a portfolio, and how sensitive is the result to the period, the costs, the parameters and stressed markets?

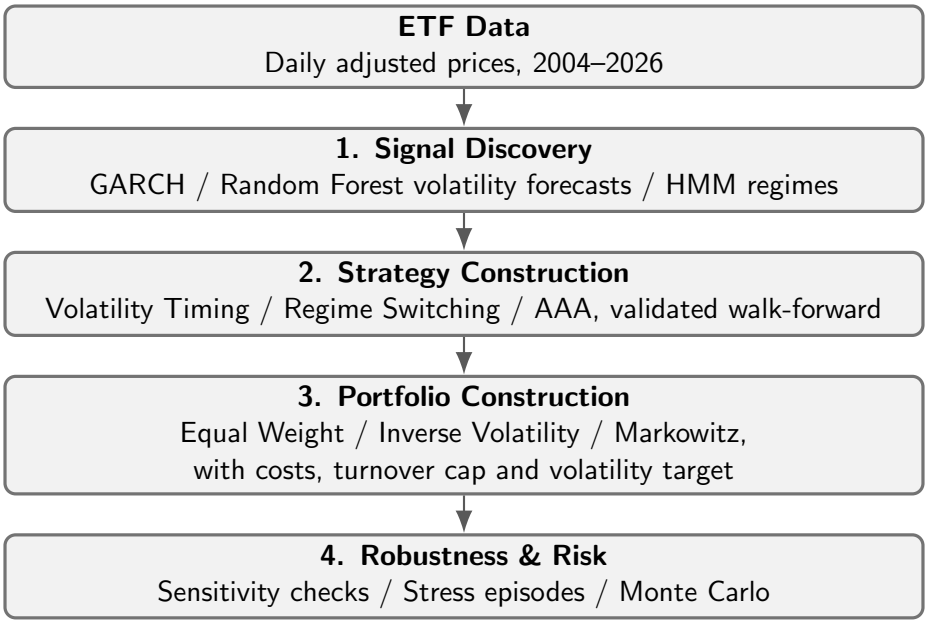


Figure 1: Project workflow.

Data. The universe is eight liquid ETFs covering U.S. equities (SPY), long-dated Treasuries (TLT), gold (GLD), oil (USO), the U.S. dollar (UUP), developed and emerging-market equities (EFA, EEM) and the euro (FXE). Adjusted close prices run from January 2004 to August 2026. GLD, FXE, USO and UUP started trading between November 2004 and March 2007; each enters the data on its first trading day, and no strategy can hold an ETF before it exists.

Conventions. Statistical models use daily log returns, $r_t = \log(S_t/S_{t-1})$; backtest accounting uses simple returns, because a weighted sum of simple returns is exactly the portfolio return. Positions are lagged one trading day before they earn returns. Each unit of turnover is charged 5 basis points of transaction cost plus 2 of slippage (7 bps in total). Portfolios use a turnover cap of 0.35 and a 10% annualized volatility target. SPY buy-and-hold and a monthly rebalanced 60/40 SPY/TLT portfolio are the benchmarks, charged the same costs. The performance and risk metrics (Sharpe, Sortino, CAGR, maximum drawdown, Calmar, volatility, VaR and Expected Shortfall) are defined in Appendix Appendix A:.

2 Signal Discovery

2.1 Volatility forecasting

The first question is whether next-period SPY risk is predictable. Two models are compared. GARCH$(1,1)$ models today's variance from yesterday's shock and yesterday's variance,

$$r_t = \sigma_t \varepsilon_t, \qquad \sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2. \tag{1}$$

A random forest (300 trees, maximum depth 4) predicts next-day realized volatility, $\hat{\sigma}_{t+1} = f(x_t)$, from return lags of 1 and 5 days, 63-day rolling mean, standard deviation, skewness and kurtosis, and the current 21-day realized volatility. Both models are estimated on the first 70% of the sample and evaluated on the last 30%, against the same target (the 21-day annualized realized volatility one day ahead) and in the same annualized units.

Model	MSE	MAE	Correlation
Random Forest	0.000255	0.008369	0.990378
GARCH$(1,1)$	0.001395	0.022807	0.945221

Table 1: Volatility forecasts on the test period. MSE and MAE are in annualized volatility units (an MAE of 0.01 is one volatility point).

Both forecasts track realized volatility closely, and the random forest has the smaller errors. The high correlations should not be over-interpreted: tomorrow's 21-day realized volatility shares 20 of its 21 days with today's, so much of the accuracy comes from volatility being persistent. The GARCH forecast is a one-day conditional volatility, which is a slightly different quantity from the 21-day target, and this also favours the random forest. In either case this is a forecast of risk, not of market direction.

2.2 Market regimes

The second question is whether SPY's history can be split into recognisable regimes. A three-state Gaussian Hidden Markov Model (HMM) is fitted to standardized daily return, 21-day realized volatility and 21-day momentum, with full covariance matrices. The hidden state follows a Markov chain, $\mathbb{P}(Z_t = j \mid Z_{t-1} = i) = A_{ij}$, and the features are Gaussian within each state.

State	Return mean	Return std.	Realized vol. mean	Days
Calm	0.0008	0.0055	0.0887	2469
Intermediate	0.0003	0.0104	0.1603	2510
Stress	-0.0004	0.0250	0.3639	700

Table 2: HMM regimes on the full sample, ordered by volatility. Return statistics are daily log returns; realized volatility is annualized (multiply by 100 for percent).

The model separates a calm state (low volatility, highest average return), an intermediate state, and a less frequent stress state with negative average return and roughly four times the volatility of the calm state. Table 2 is descriptive and fitted to the full sample; the trading rule in Section 3 refits the HMM on training data only.

3 Strategy Construction

All strategies use the same backtest engine. If w_t are the weights chosen at time t and $\tau_t = \sum_i |w_{i,t} - w_{i,t-1}|$ is turnover, the net return is

$$r_{t+1}^{\text{net}} = w_t^\top r_{t+1} - \tau_t \frac{c}{10{,}000}, \qquad c = 7 \text{ bps.} \tag{2}$$

Volatility timing (VolTiming) scales SPY exposure inversely to the random-forest volatility forecast, $w_t \propto 1/\hat{\sigma}_{t+1}$, normalized by its expanding average and clipped between 0 and 1.5. It holds less SPY when forecast risk is high and more when it is low.

Regime switching (RegimeHMM) maps the HMM state to SPY exposure: 1.00 in the bull state, 0.35 in the sideways state and 0 in the crisis state. The states are labelled by their average return on the training window; the exposure levels are fixed policy choices, not fitted parameters.

Adaptive asset allocation (AAA) is the only strategy that uses the whole universe. At each month-end it computes each ETF's momentum over a lookback of L days,

$$M_{i,t} = \frac{P_{i,t}}{P_{i,t-L}} - 1, \tag{3}$$

keeps the set $\mathcal{S}_t$ of the N ETFs with the highest momentum, and weights them by inverse 63-day volatility $\hat{\sigma}_{i,t}$ (floored at 5%),

$$\tilde{w}_{i,t} \;=\; \mathbf{1}\{i \in \mathcal{S}_t\} \frac{1/\hat{\sigma}_{i,t}}{\sum_{j \in \mathcal{S}_t} 1/\hat{\sigma}_{j,t}}, \qquad \mathbf{1}\{i \in \mathcal{S}_t\} = \begin{cases} 1, & \text{if ETF } i \text{ is selected,} \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

The indicator $\mathbf{1}\{\cdot\}$ simply switches an ETF on or off, so unselected ETFs get zero weight and the selected weights sum to one. Any selected ETF whose own momentum is negative has its

weight moved to TLT, the only bond fund in the universe (or to cash if TLT is not yet available). The rule is long-only, rebalances monthly and estimates nothing: every weight is a fixed function of past prices.

3.1 Walk-forward validation

Every strategy and portfolio number in this document is out of sample. Training starts with two years of data (504 days) and grows by one month (21 days) at each fold; positions are recorded only on the month that follows each training window. This gives 241 folds and $5{,}061$ out-of-sample days from July 2006 to August 2026.

The random forest and the HMM are re-estimated on each training window. AAA has no model to refit, but its lookback and selection size $\theta = (L, N)$ are chosen on each training window by net Sharpe ratio,

$$\theta_k^\star = \arg\max_{\theta \in \{63,126,252\} \times \{3,4,5\}} \mathrm{Sharpe}\big(r_\theta^{\mathrm{net}} \text{ on training window } k\big), \tag{5}$$

and then used for the next month. In practice the choice alternated between $(126, 4)$ and $(63, 5)$ (Table B8). Separate checks confirmed that AAA's weights do not change when future data are removed, that every parameter choice can be reproduced from its training window alone, and that no ETF is held before it started trading.

3.2 Strategy results

Metric	VolTiming	RegimeHMM	AAA	SPY	60/40
Sharpe ratio	0.81	0.75	0.83	0.65	0.82
Sortino ratio	1.05	0.83	1.10	0.80	1.06
CAGR (%)	10.68	6.18	8.59	11.35	9.00
Max drawdown (%)	-30.18	-14.95	-17.32	-55.19	-29.92
Calmar ratio	0.35	0.41	0.50	0.21	0.30
Volatility (%)	13.69	8.52	10.61	19.38	11.33
VaR_{95} (%)	1.45	0.83	1.06	1.79	1.06
ES_{95} (%)	2.12	1.40	1.61	2.98	1.68
Turnover (x/yr)	6.81	11.42	6.78	–	–

Table 3: Strategy performance, walk-forward out of sample, net of costs, July 2006 to August 2026.

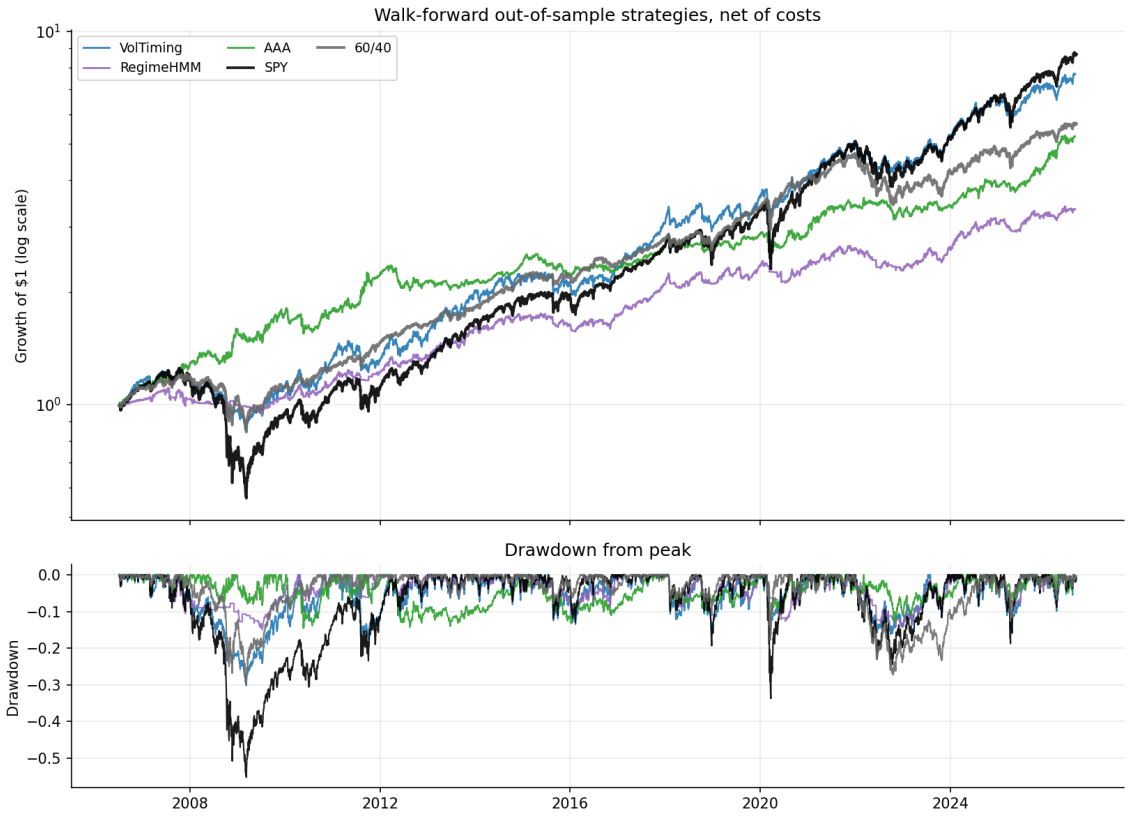


Figure 2: Out-of-sample equity and drawdown curves, net of costs.

All three strategies have a higher Sharpe ratio than SPY buy-and-hold and much smaller drawdowns, but lower growth: each spends part of the time out of equities. Their Sharpe ratios (0.75 to 0.83) are close to each other and to the 60/40 benchmark (0.82), so the ranking between them is not meaningful on its own. The clearer differences are in drawdown. Volatility timing still lost 30%, because a volatility forecast reacts after a crash has started. Regime switching had the smallest drawdown (-15%) and the lowest growth. AAA's drawdown (-17%) was a third of SPY's, mostly because it moved into Treasuries and gold in 2008.

4 Portfolio Construction

The strategies are combined in two menus that differ by one strategy, so the effect of adding AAA can be measured directly: **Orig2** = {VolTiming, RegimeHMM} and **Core3** = Orig2 + AAA. Each menu is combined in four ways: equal weight; inverse volatility, $w_m \propto 1/\hat{\sigma}_m$ using 63-day strategy volatility; long-only maximum-Sharpe Markowitz,

$$\max_{w} \frac{w^\top \hat{\mu}}{\sqrt{w^\top \hat{\Sigma} w}}, \qquad \textstyle\sum_m w_m = 1,\ 0 \le w_m \le 1, \tag{6}$$

with 252-day estimates; and a regularized Markowitz that subtracts $0.25 \sum_m w_m^2$ from the objective to discourage concentration. Every portfolio then goes through the same steps: a turnover cap of 0.35, costs of 7 bps charged a second time at the portfolio level, and a volatility target that scales the book by $\lambda_t = 0.10/\hat{\sigma}_t$ (clipped to $[0, 2]$, using the previous day's estimate).

Diversification is the main reason AAA can help. VolTiming and RegimeHMM both trade SPY and their returns correlate at 0.84; AAA correlates about 0.4 with each (Table 4).

Strategy	VolTiming	RegimeHMM	AAA
VolTiming	1.00	0.84	0.41
RegimeHMM	0.84	1.00	0.43
AAA	0.41	0.43	1.00

Table 4: Correlation of out-of-sample strategy returns.

Menu	Equal weight	Inverse vol	Markowitz	Markowitz regularized
Orig2	0.79	0.77	0.72	0.72
Core3	0.96	0.88	0.82	0.82

Table 5: Net Sharpe ratio by menu and weighting method, walk-forward and after all costs.

Menu	Equal weight	Inverse vol	Markowitz	Markowitz regularized
Orig2	-23.8	-23.5	-18.0	-21.4
Core3	-16.0	-16.8	-21.1	-21.2

Table 6: Maximum drawdown (%) for the same portfolios.

Metric	Orig2 inv-vol	Orig2 equal	Core3 inv-vol	Core3 equal	Core3 Mkw. reg.	SPY	60/40
Sharpe ratio	0.77	0.79	0.88	0.96	0.82	0.64	0.80
Sortino ratio	0.94	0.98	1.10	1.22	1.05	0.79	1.04
CAGR (%)	7.79	8.00	9.09	10.03	8.40	11.22	8.80
Max drawdown (%)	-23.52	-23.76	-16.79	-15.97	-21.19	-55.19	-29.92
Calmar ratio	0.33	0.34	0.54	0.63	0.40	0.20	0.29
Volatility (%)	10.46	10.47	10.50	10.59	10.58	19.47	11.38
VaR_{95} (%)	1.03	1.05	1.05	1.07	1.09	1.80	1.07
ES_{95} (%)	1.68	1.65	1.66	1.63	1.62	2.99	1.69

Table 7: Key portfolios in detail, with SPY and 60/40 over the same window (slightly shorter than the strategy window because of the portfolio estimation windows).

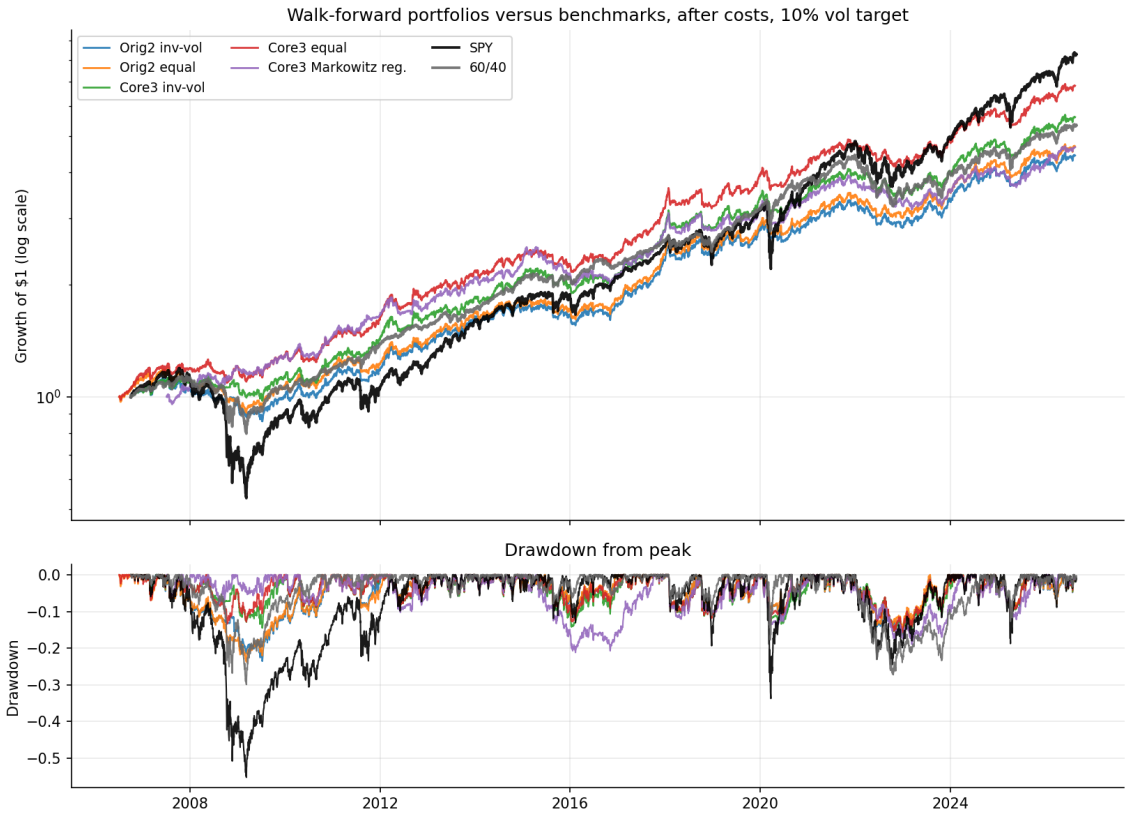


Figure 3: Portfolio equity and drawdown curves after turnover cap, costs and volatility targeting.

Three observations follow. First, Core3 has a higher Sharpe ratio than Orig2 under every weighting method, by 0.10 to 0.17. Its maximum drawdown is clearly smaller under equal and inverse-volatility weighting (-16.0% and -16.8% against about -23.5%), but not under the two Markowitz methods. Second, the simple methods did at least as well as optimization: for Core3, equal weight reached 0.96, inverse volatility 0.88 and both Markowitz versions 0.82. A likely reason is that expected returns estimated from a year of data are noisy. Third, compared with SPY, Core3 (inverse volatility) had a higher Sharpe ratio (0.88 against 0.64) and a much smaller maximum drawdown (-16.8% against -55.2%), but lower annual growth (9.1% against 11.2%). Inverse volatility is used as the headline portfolio below; its gap to equal weighting is small.

5 Robustness and Risk

5.1 Sensitivity checks

The supporting tables are in Appendix Appendix B:; the main points are:

- **Sub-periods.** AAA has a positive Sharpe ratio in all five sub-periods, with its best result in 2006–09. Core3 is positive in every sub-period and at or above SPY in each, but below 60/40 in 2010–21, when bonds and equities both performed well.
- **Costs.** At 30 bps per unit of turnover (more than four times the base assumption), AAA still has a Sharpe ratio of 0.68, slightly above SPY's 0.65. Core3, which pays costs at both the strategy and the portfolio level, falls to 0.57, below SPY.
- **Parameters.** In a grid of 20 fixed AAA settings, 15 have a Sharpe ratio above SPY's. This grid is scored on the same out-of-sample window, so its best cell (0.95) is optimistic; the walk-forward figure of 0.83 is the one to rely on. Using GLD instead of TLT as the defensive asset changes the Sharpe ratio by less than 0.01.
- **Overlay.** The turnover cap never binds, so it has no effect. The volatility target mainly trades return against drawdown: an 8% target gives a Sharpe ratio of 0.89 and a -13.6% drawdown, a 12% target 0.87 and -19.9%.

5.2 Historical stress episodes

Each strategy's out-of-sample daily returns are cut to the dates of nine historical market episodes, and the compounded return over each window is reported. No shocks are simulated; this shows what the strategies actually did in those periods.

Episode	VolTiming	RegimeHMM	AAA	Core3 inv-vol	SPY	60/40
2008 global financial crisis	-28.8	-8.8	24.2	-9.9	-54.0	-28.6
2010 flash crash / euro scare	-11.5	-6.6	-7.4	-7.9	-14.7	-3.4
2011 US downgrade / euro crisis	-13.2	-3.0	21.9	1.3	-16.3	1.8
2015 China deval / Aug flash crash	-9.2	-5.6	-0.7	-4.9	-8.5	-4.5
2018 Feb VIX spike	-6.7	-5.0	-5.3	-6.8	-4.1	-4.0
2018 Q4 rate-hike selloff	-12.5	-6.5	-1.3	-8.7	-13.5	-6.5
2020 COVID crash	-10.5	-5.3	-6.1	-8.8	-17.2	-3.1
2022 inflation / hiking cycle	-11.8	-10.7	-6.3	-12.3	-17.7	-24.0
2025 tariff shock	-4.0	-4.7	-3.3	-4.8	-0.5	-2.3

Table 8: Total return (%) through each stress episode, walk-forward and net of costs.

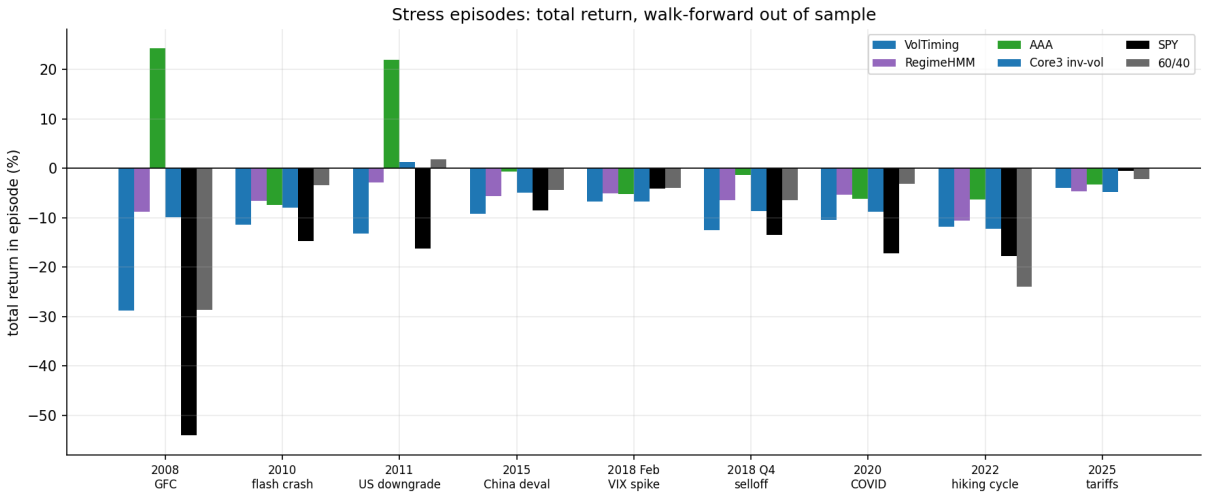


Figure 4: Total return in each stress episode.

In the 2008 crisis SPY lost 54% and 60/40 lost 29%, while Core3 lost 10% and AAA gained 24%. AAA also gained in 2011. In most of the other episodes the strategies lost less than SPY. The exceptions are short, sharp shocks such as February 2018 and the 2025 tariff episode, where the strategies lost slightly more than SPY because monthly or lagged signals cannot react within days. 2022 was Core3's weakest episode (-12.3%): equities and bonds fell together, and Treasuries are the only defensive asset in the universe.

5.3 Monte Carlo

A historical backtest is a single path. To see a range of plausible one-year outcomes, $5{,}000$ alternative years (252 days each) are built by resampling each portfolio's own out-of-sample daily returns in blocks of consecutive days (stationary block bootstrap, average block length 21 days). Blocks rather than single days are used so that calm and volatile periods stay clustered as they are in real markets.

Portfolio	Median 1y (%)	5th pct 1y (%)	95th pct 1y (%)	P(loss) (%)	MC ES_{95} (%)	Median max DD (%)
Orig2 inv-vol	7.73	-8.43	26.57	22.20	11.99	-7.94
Orig2 equal	8.12	-8.23	26.12	21.82	11.91	-8.07
Core3 inv-vol	9.07	-7.32	28.04	18.94	11.25	-7.80
Core3 equal	10.06	-7.06	28.59	17.02	10.72	-7.68
Core3 Markowitz reg.	8.66	-8.27	27.24	21.42	12.26	-7.84
AAA	8.45	-6.91	26.30	18.00	10.68	-7.26
SPY	12.89	-17.65	42.02	23.36	25.43	-13.25
60/40	9.62	-9.05	27.00	18.64	13.79	-8.06

Table 9: Distribution of one-year outcomes across 5,000 block-bootstrap paths.

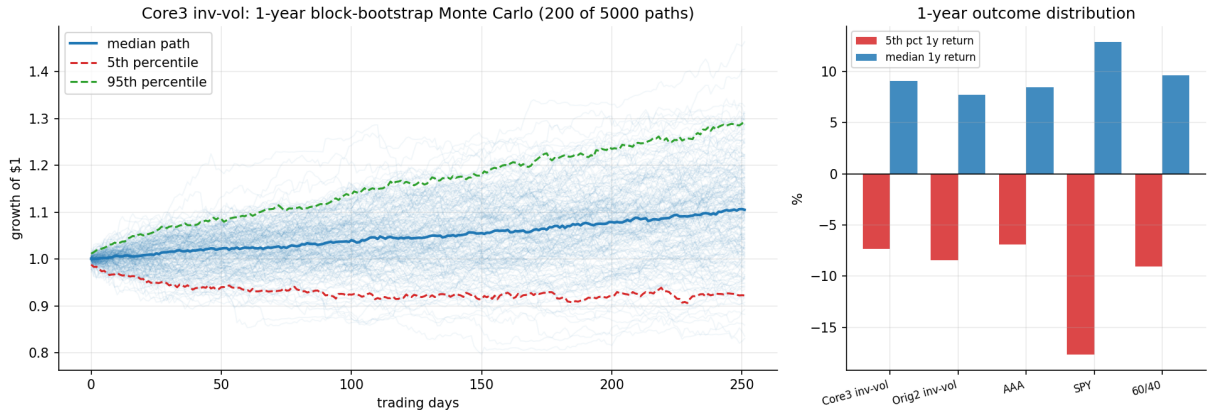


Figure 5: Left: 200 of the 5,000 simulated one-year paths for Core3. Right: median and 5th-percentile one-year returns.

Core3's median simulated year returned 9.1% against 12.9% for SPY, its 5th-percentile year lost 7.3% against 17.7%, and the probability of a losing year was 19% against 23%. This matches the historical record: less upside and less downside than SPY. Because the paths are built only from past returns, the simulation cannot produce a type of event that is not already in the sample.

6 Limitations

- The results come from one historical sample of about 20 out-of-sample years and eight ETFs. They are evidence, not a guarantee of future performance.
- Costs are a fixed 7 bps per unit of turnover. Market impact, wider spreads in stressed markets, taxes and capacity are not modelled.
- Walk-forward selection removes hindsight about parameters, but not about the choice of strategies. AAA is a well-known approach, chosen with knowledge of the momentum literature.
- Treasuries are the only defensive asset, so periods like 2022, when stocks and bonds fall together, are a structural weakness.
- Several differences in the tables (for example between equal and inverse-volatility weighting, or between the three strategies' Sharpe ratios) are small relative to the uncertainty in a Sharpe ratio estimated over 20 years.

7 Summary & Conclusion

The project takes ideas from signal research through to a costed, walk-forward tested portfolio. Volatility and market regimes could be modelled usefully, and strategies built on them reduced

drawdowns relative to holding SPY. A cross-asset momentum strategy (AAA) behaved differently from the two SPY-based strategies, and adding it to the portfolio improved the risk-adjusted return and reduced the drawdown under the simple weighting methods. Simple weighting did at least as well as optimization. The robustness checks, stress episodes and Monte Carlo point the same way: the portfolios gave up some return relative to SPY in exchange for considerably smaller losses in bad periods, with 2022 and sudden short shocks as the main weaknesses.

Appendix A: Performance and Risk Metrics

Let $r_1, \ldots, r_T$ be daily returns, $A = 252$ the annualization factor, $\bar{r}$ the sample mean and s_r the sample standard deviation.

Sharpe ratio: $\mathrm{Sharpe} = \sqrt{A}\,\bar{r}/s_r$. The risk-free rate is omitted because all return streams are compared on the same basis.

Sortino ratio: $\mathrm{Sortino} = \sqrt{A}\,\bar{r}/s_r^-$, where s_r^- is the standard deviation of the negative returns only, so upside variation is not penalized.

Equity curve and CAGR: $E_t = \prod_{u \le t}(1 + r_u)$ with $E_0 = 1$; if $Y = T/A$ is the length in years, $\mathrm{CAGR} = E_T^{1/Y} - 1$.

Maximum drawdown: the worst fall from a previous peak,

$$\mathrm{MDD} = \min_{t \le T} \left(\frac{E_t}{\max_{u \le t} E_u} - 1 \right).$$

Calmar ratio: $\mathrm{CAGR}/|\mathrm{MDD}|$, growth per unit of worst loss.

Annualized volatility: $\sigma_{\mathrm{ann}} = s_r\sqrt{A}$.

Value at Risk: with $\hat{q}_{0.05}$ the empirical 5th percentile of daily returns, $\mathrm{VaR}_{95} = -\hat{q}_{0.05}$, reported as a positive loss. A daily VaR_{95} of 1% means that on the worst 5% of days the loss was at least about 1%.

Expected Shortfall: the average loss on those worst days, $\mathrm{ES}_{95} = -\frac{1}{|\mathcal{T}|}\sum_{t \in \mathcal{T}} r_t$ with $\mathcal{T} = \{t : r_t \le \hat{q}_{0.05}\}$. It is at least as large as VaR and describes how bad the tail is, not just where it starts.

In the Monte Carlo table the same quantities are computed on simulated one-year returns rather than daily returns.

Appendix B: Supplementary Results

Strategy	2006-09	2010-13	2014-17	2018-21	2022-26
VolTiming	0.29	1.06	0.85	0.97	0.78
RegimeHMM	0.20	1.28	0.83	0.67	0.66
AAA	1.41	0.51	0.75	0.62	0.95
SPY	0.07	0.96	0.99	0.88	0.76
60/40	0.27	1.56	1.52	1.19	0.38

Table B1: Net Sharpe ratio by sub-period, strategies.

Portfolio	2006-09	2010-13	2014-17	2018-21	2022-26
Orig2 inv-vol	0.03	1.25	0.89	0.88	0.66
Orig2 equal	0.20	1.18	0.90	0.91	0.68
Core3 inv-vol	0.41	1.23	1.03	0.88	0.77
Core3 equal	0.72	1.30	0.98	0.98	0.79
Core3 Markowitz reg.	0.98	1.22	0.72	0.78	0.48
SPY	0.02	0.96	0.99	0.88	0.76
60/40	0.17	1.56	1.52	1.19	0.38

Table B2: Net Sharpe ratio by sub-period, portfolios.

Strategy	0bps	5bps	7bps	10bps	15bps	20bps	30bps	assumed_bps
VolTiming	0.85	0.82	0.81	0.79	0.77	0.74	0.69	7.00
RegimeHMM	0.84	0.77	0.75	0.71	0.64	0.57	0.43	7.00
AAA	0.88	0.84	0.83	0.81	0.78	0.75	0.68	7.00

Table B3: Net Sharpe ratio as the cost per unit of turnover rises, strategies.

Cost	Orig2 inv-vol	Core3 inv-vol
0bps	0.85	0.98
5bps	0.79	0.91
7bps	0.77	0.88
10bps	0.74	0.84
15bps	0.68	0.77
20bps	0.62	0.71
30bps	0.51	0.57

Table B4: Net Sharpe ratio as costs rise, inverse-volatility portfolios (costs charged at strategy and portfolio level).

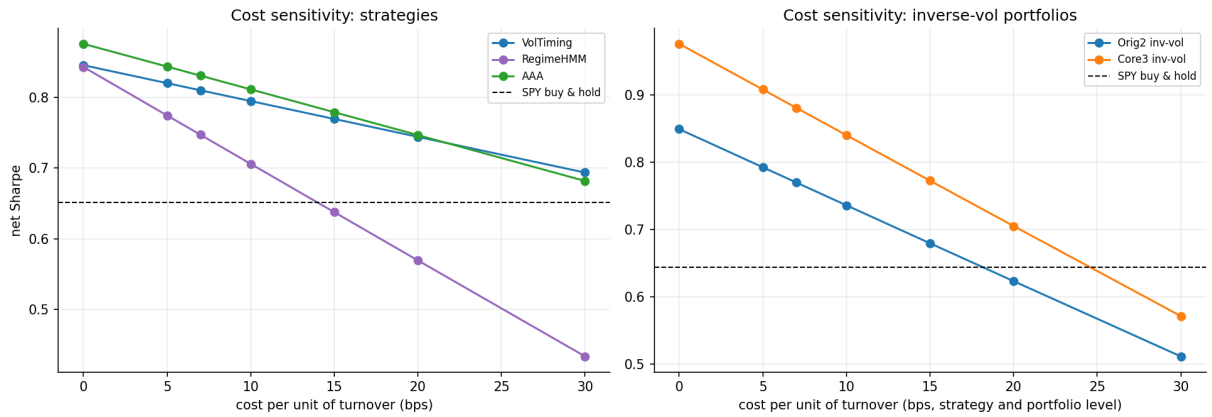


Figure B1: Cost sensitivity. The dashed line is SPY buy-and-hold.

Selection	lb 63	lb 126	lb 189	lb 252
top 2	0.54	0.81	0.69	0.46
top 3	0.82	0.85	0.65	0.62
top 4	0.82	0.95	0.73	0.70
top 5	0.81	0.80	0.74	0.71
top 6	0.76	0.71	0.67	0.61

Table B5: AAA net Sharpe for fixed settings: number of ETFs held (rows) and momentum lookback in days (columns), scored on the out-of-sample window.

Vol. window	defensive TLT	defensive GLD
vol 21	0.95	0.95
vol 42	0.95	0.95
vol 63	0.95	0.95
vol 126	0.95	0.95

Table B6: AAA net Sharpe by volatility window and defensive asset.

Variant	sharpe	max_drawdown
VolTiming leverage cap=1.0	0.85	-0.30
VolTiming leverage cap=1.25	0.83	-0.30
VolTiming leverage cap=1.5	0.81	-0.30
VolTiming leverage cap=2.0	0.81	-0.30
RegimeHMM sideways weight=0.0	0.67	-0.18
RegimeHMM sideways weight=0.2	0.82	-0.13
RegimeHMM sideways weight=0.35	0.87	-0.14
RegimeHMM sideways weight=0.5	0.88	-0.18
RegimeHMM sideways weight=0.7	0.87	-0.24

Table B7: VolTiming and RegimeHMM: Sharpe ratio and maximum drawdown (as a fraction) when the single free parameter of each rule is varied.

Parameter set	share_of_folds
lb126_top4	0.502
lb63_top5	0.494
lb63_top4	0.004

Table B8: AAA parameter set selected by the walk-forward, as a share of the 241 folds.

Turnover cap	vol 8% Sharpe	vol 8% MaxDD (%)	vol 10% Sharpe	vol 10% MaxDD (%)	vol 12% Sharpe	vol 12% MaxDD (%)
cap 0.10	0.89	-13.60	0.88	-16.78	0.87	-19.91
cap 0.20	0.89	-13.61	0.88	-16.79	0.87	-19.93
cap 0.35	0.89	-13.61	0.88	-16.79	0.87	-19.93
cap 0.50	0.89	-13.61	0.88	-16.79	0.87	-19.93
cap 1.00	0.89	-13.61	0.88	-16.79	0.87	-19.93

Table B9: Core3 (inverse volatility) under different turnover caps and volatility targets.

Rank	Peak	Trough	Recovery	Depth (%)	Days to trough	Days to recover
1	2021-11-08	2022-11-09	2024-01-24	-16.8	253	301.0
2	2008-05-19	2009-07-08	2010-01-04	-14.5	286	124.0
3	2015-04-24	2016-01-11	2017-02-13	-14.2	180	275.0
4	2020-01-17	2020-03-18	2021-01-07	-13.5	41	204.0
5	2018-01-26	2018-12-24	2019-04-04	-11.2	229	69.0

Table B10: The five deepest Core3 drawdowns, with trading days to the trough and to recovery.

Strategy	Bull	Sideways	Crisis
AAA	1.45	0.67	0.43
Core3 inv-vol	2.16	0.18	-0.62
RegimeHMM	2.35	-0.07	-1.36
SPY	2.51	0.53	-0.07
VolTiming	2.30	0.11	-0.38

Table B11: Sharpe ratio within each HMM regime. The labels are fitted on the full sample, so this is a description, not a tradable result.

Portfolio	AAA	RegimeHMM	VolTiming
Orig2 Inverse vol	–	62.7	37.3
Orig2 Markowitz regularized	–	48.9	51.1
Core3 Inverse vol	35.2	41.1	23.7
Core3 Markowitz regularized	38.4	30.3	31.3

Table B12: Average strategy weights (%) in each portfolio after the turnover cap.

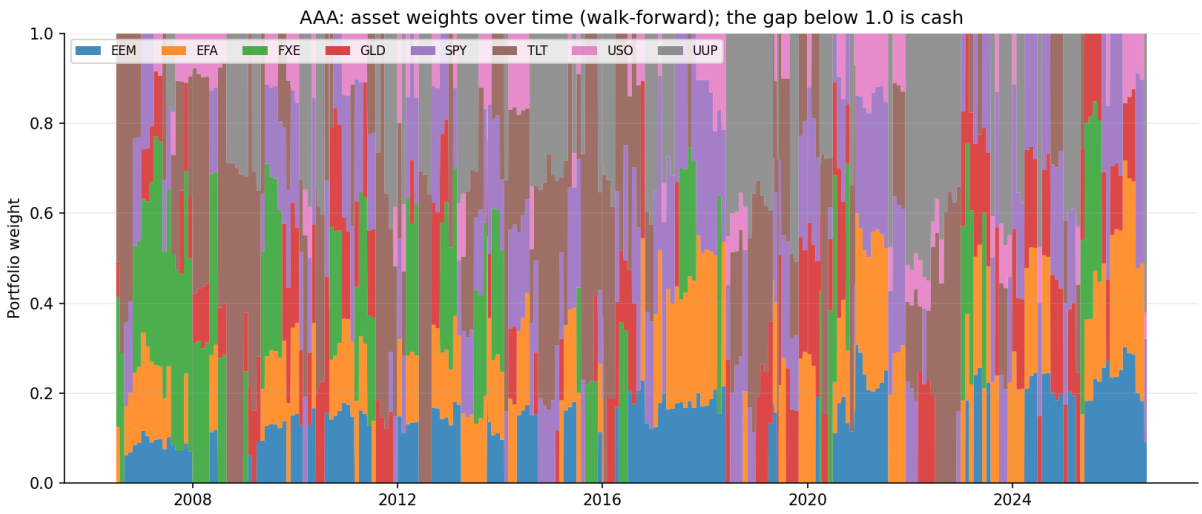


Figure B2: AAA weights over time. Large TLT allocations show the negative-momentum rule moving weight to Treasuries.

Appendix C: Reproducing the Results

The full pipeline (data download, walk-forward strategies, portfolios, robustness and risk analysis) runs with `python run_research.py`; see the repository README for details.